

# CEL: Comprehensive Counterfactual Explanations Library and Benchmark

Oleksii Furman · Łukasz Lenkiewicz · Marcel Musiałek · Maciej Zięba

Wrocław University of Science and Technology, Wrocław, Poland

## What is a counterfactual explanation?

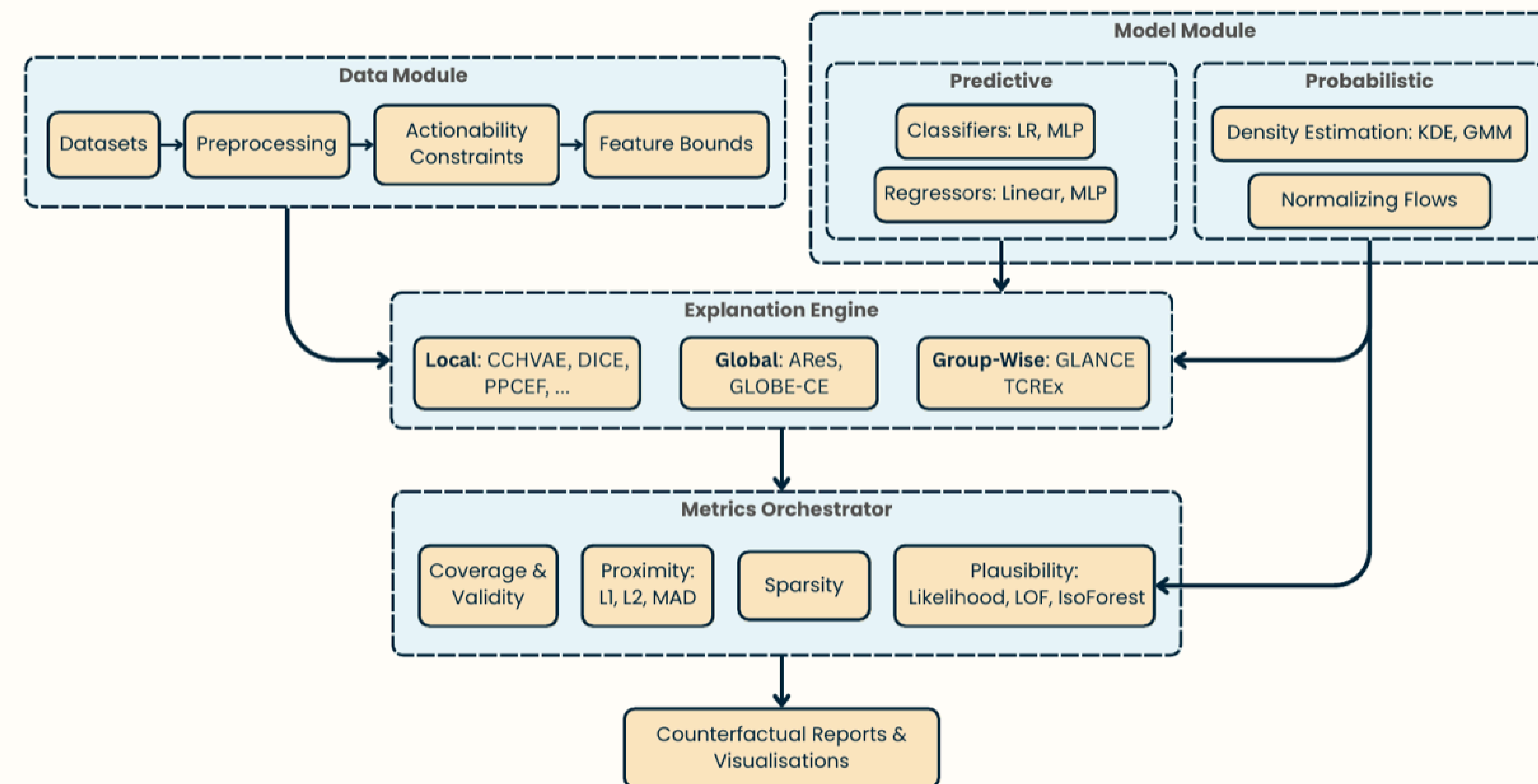
A small input change that flips a model's prediction.

### LOAN APPLICATION EXAMPLE

| Feature        | Original  | Counterfactual |
|----------------|-----------|----------------|
| Age            | 35 years  | 35 years       |
| Employment     | Full-time | Full-time      |
| Credit history | 6 years   | 6 years        |
| Loan amount    | €15,000   | €15,000        |
| Monthly income | €3,000    | €3,500         |
| Monthly debt   | €1,200    | €700           |

Prediction **Declined** **Approved**

Only 2 of 6 features change.



## 18 datasets

Adult Census · Audit · Bank Marketing · Blobs · Credit Default · Digits · German Credit · Give Me Some Credit · HELOC · Law · Lending Club · Moons · Wine · Synthetic · Concrete · Diabetes · Yacht · SCM20D

## 14 methods

**Local**  
Wachter · Artelt · DiCE · CCHVAE · PPCEF · CEM · CEGP · CADEX · SACE · CEARM

**Global**  
ARes · GLOBE-CE

**Group-wise**  
GLANCE · T-CREX

## 2 backbones / task

**Classification**  
Multi-Layer Perceptrons (MLP) · Logistic Regression

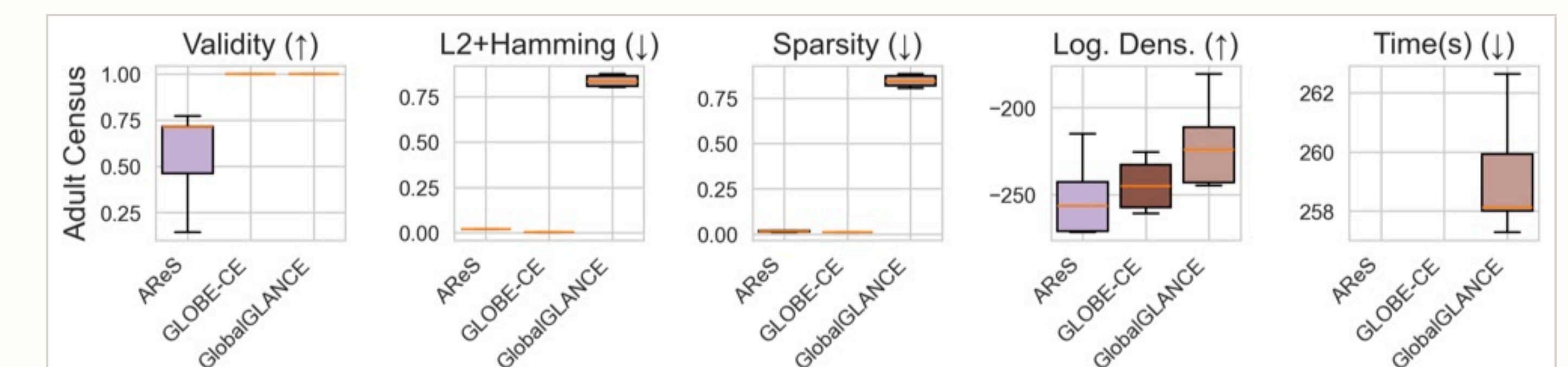
**Continuous targets**  
Linear Regression · MLP Regressor

## 9 classification metrics

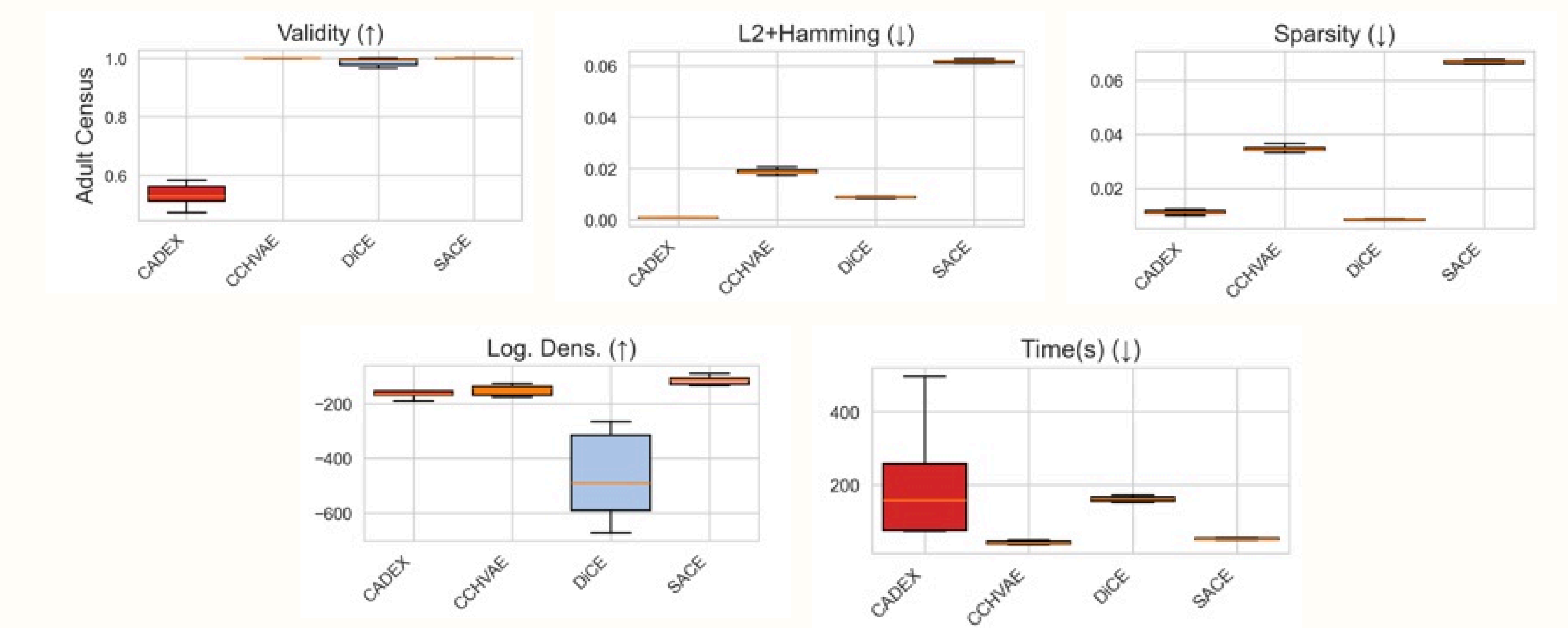
Coverage · Validity · Sparsity · Prob. plausibility · Log-density · LOF · Isolation Forest · Proximity (L2 / L2-Hamming) · Runtime

## Results

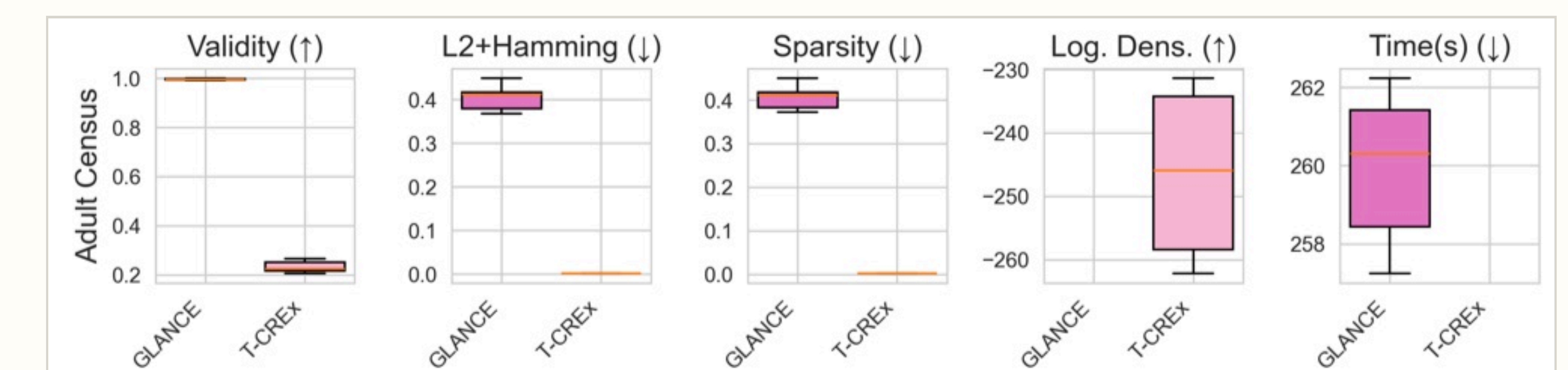
### Global: validity and failures



### Local: no single winner



### Group-wise: reach versus change



## Three contributions

**CONTROL**  
Controlled evaluation protocol

**COVER**  
Broad benchmark across CE paradigms

**EXTEND** Extensible open-source benchmark workbench



Code & project